Simulation Study Plan

shg_mode_survey · geometry × material design matrix, agreed 2026-10-12

This documents the campaign before any of it runs -- ten geometry/material combinations (MATRIX, next page), grouped into four underlying cross-section topologies (one page each, with a real diagram where the family already exists, a schematic where it doesn't yet).

Built from the generalized-geometry sketches shared 2026-10-12: a 3-layer stack (A = base film/rib, C = core ridge, B = loading/shell region), where whether B just caps C's top or wraps around its sides, and how large B is, span both the guided-mode families already in this repo (ridge, loaded, the corrected hybrid) and the not-yet-built Cerenkov case in one consistent vocabulary -- not four unrelated structures.

READING THE MATRIX

status: ✓ ready to run now, with existing materials data.
○ blocked on a materials gap (d33(Sc), GaAlN n/k/d33, or an anisotropic-substrate decision) -- see the Input Assumptions report for what's known and what isn't.
h_A: the base-layer thickness, free 0 (pure ridge/no embedding base) up to a full film (rib/loaded) -- only wired up in the `loaded` family today; adding it to the embedded topologies (rows 6-9) is new geometry work, not yet done (marked TBD*).

STANDING ASSUMPTION FOR EVERY NEW RUN FROM HERE ON

AlN's own measured SH-band absorption (AM187-LT6 k_e) was explored and REJECTED (it implied ~100,000 dB/cm, checked against real device output and found 3-4 orders of magnitude too high). Every row below should instead launch with geom.params['sh_absorption_calibrated'] = True -- a flat, empirically-calibrated 100 dB/cm SH-band addition, confinement-weighted, layered on top of native scattering (unaffected). See Input Assumptions report section 3b and PLAN.md's 2026-10-12 entries for the full derivation.

Not a commitment to run all ten at once -- user direction, 2026-10-12: check in before each batch.

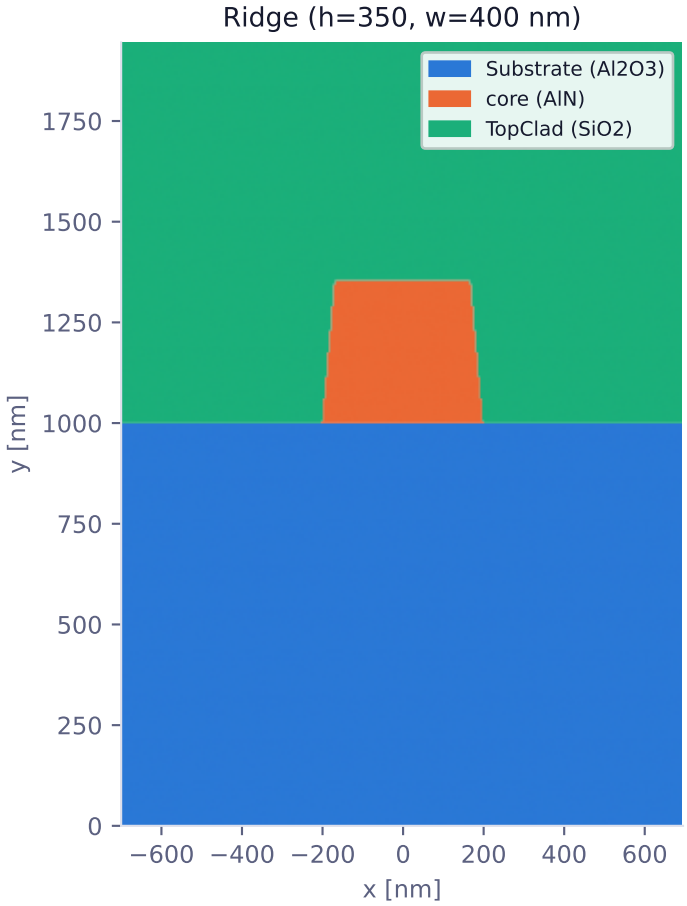
Design matrix

10 rows, grouped by topology — see the following pages for diagrams of each

#		TOPOLOGY	H_A	CORE (C)	SHELL/OUTER (B)	SUBSTRATE	PURPOSE	STATUS
1	✓	Ridge	n/a (no A layer)	AlN	—	Al2O3	baseline	done (`ridge`); loss comparison done
2	✓	Ridge	n/a	ScAlN	—	Al2O3	does Sc-enhanced d33 alone beat AlN, no loading tricks	ready — d33(Sc) set, AT RISK
3	✓	Rib-loaded	0 → film thickness	AlN	TiO2	Al2O3	baseline	done — found bad (SH-absorbing)
4	✓	Rib-loaded	0 → film thickness	AlN	SiN	Al2O3	SiN strip instead of TiO2	ready (`loaded_sin`), not yet run
5	✓	Rib-loaded	0 → film thickness	AlN	ScAlN	Al2O3	nonlinear-active loading strip — new idea	ready — d33(Sc) set, AT RISK
6	✓	Embedded (guided)	0 → TBD*	AlN	SiN	Al2O3	corrected `hybrid` — Honda et al. mechanism	ready, re-run pending
7	✓	Embedded (guided)	0 → TBD*	AlN	ScAlN	Al2O3	both-active variant	ready — d33(Sc) set, AT RISK
8	○	Embedded (Cerenkov)	0 → TBD*	ScAlN	AlN (blanket)	Al2O3 or AlN	buried high-index channel radiating into thick AlN	d33 set; needs embedded-Cerenkov geometry (not built)
9	○	Embedded (Cerenkov)	0 → TBD*	GaAlN	AlN (blanket)	Al2O3 or AlN	same, GaAlN channel	need full GaAlN n/k/d33

Topology 1: Ridge

real cross-section, from geometry.py



MATRIX ROWS

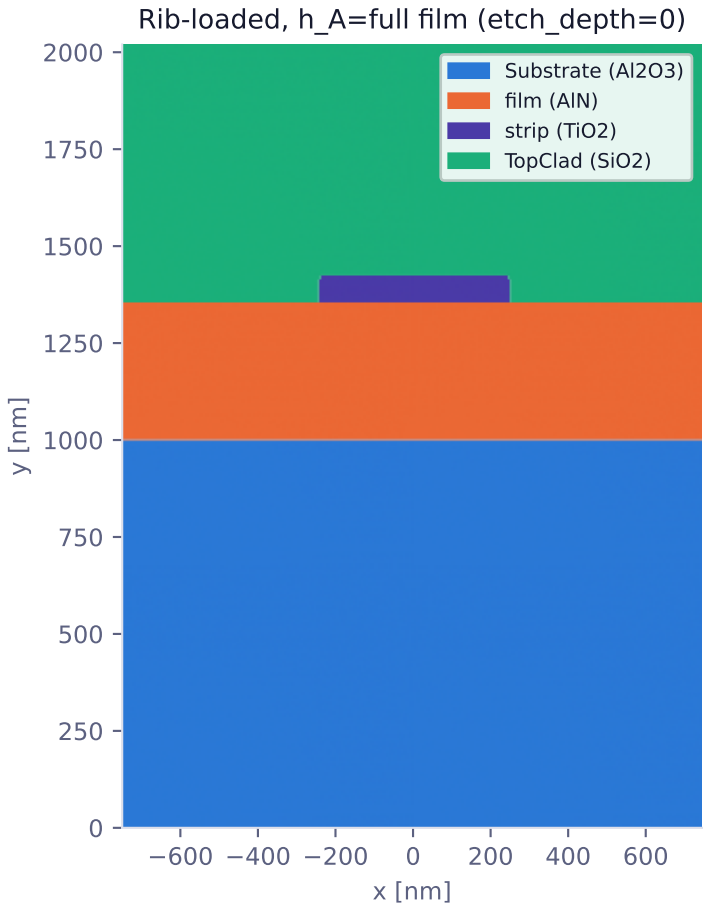
✓ **#1** AlN / — — baseline

✓ **#2** ScAlN / — — does Sc-enhanced d33 alone beat AlN, no loading tricks

Bare AlN (or ScAlN) core directly on substrate -- no A layer, no B/shell at all. The simplest structure in the matrix, and still the best-surviving one under real (calibrated) SH loss, confirmed directly (aln_loss_report.pdf): median calibrated peak efficiency ~41,000x above `loaded`'s, best-to-best ~4,700x -- `loaded`'s TiO2 strip, already lossy at SH before any of this, compounds with the calibrated addition rather than being masked by it. Row 2 asks whether Sc-doping the core alone (no shell, no loading) can beat even this -- ready to run, using the 32%-Sc working d33 (section 6b of Input Assumptions, AT RISK).
SWEPT RANGE (row 1, already done): h_core 300-400nm x w_core 300-800nm (trial_grid/ wide_window) -- best found: h=350, w=500. Row 2 (ScAlN) not yet run; same range proposed.

Topology 2: Rib-loaded (cap-only)

real cross-section, from geometry.py



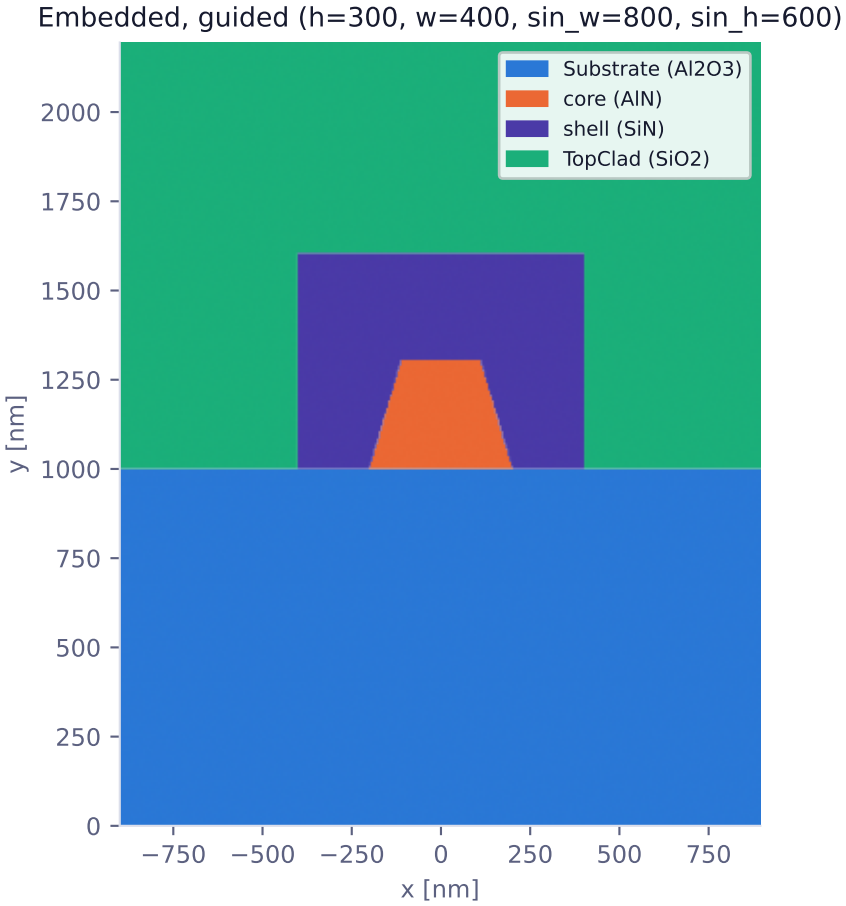
MATRIX ROWS

- ✓ **#3** AlN / TiO2 — baseline
- ✓ **#4** AlN / SiN — SiN strip instead of TiO2
- ✓ **#5** AlN / ScAlN — nonlinear-active loading strip — new idea

A (AlN film/rib) with a narrower loading strip B capping only its top (w_B <= w_C in the generalized sketch's terms) -- the existing `loaded` family. h_A ranges from a full, unetched film (shown here, etch_depth=0 -- the 'loaded waveguide' extreme) down to a fully etched ridge with just the strip's own confinement left. Row 3 (TiO2) already ran and was found bad -- TiO2 absorbs strongly at the SH wavelength, collapsing real efficiency ~40,000x below ridge's. Row 4 (SiN) swaps the strip material; row 5 (ScAlN strip) is ready to run, using the 32%-Sc working d33 (section 6b of Input Assumptions, AT RISK).
SWEPT RANGE: t_film {354,600}nm x h_s {70,150}nm x w_s {500,1000}nm, etch_depth=0 -- the same 8-point grid for rows 3/4/5 (`survey_config_loaded.py` / `_loaded_sin.py`), kept identical so a material swap is the only variable between them.

Topology 3: Embedded (guided)

real cross-section, from geometry.py



MATRIX ROWS

✓ **#6** AlN / SiN — corrected `hybrid` — Honda et al. mechanism

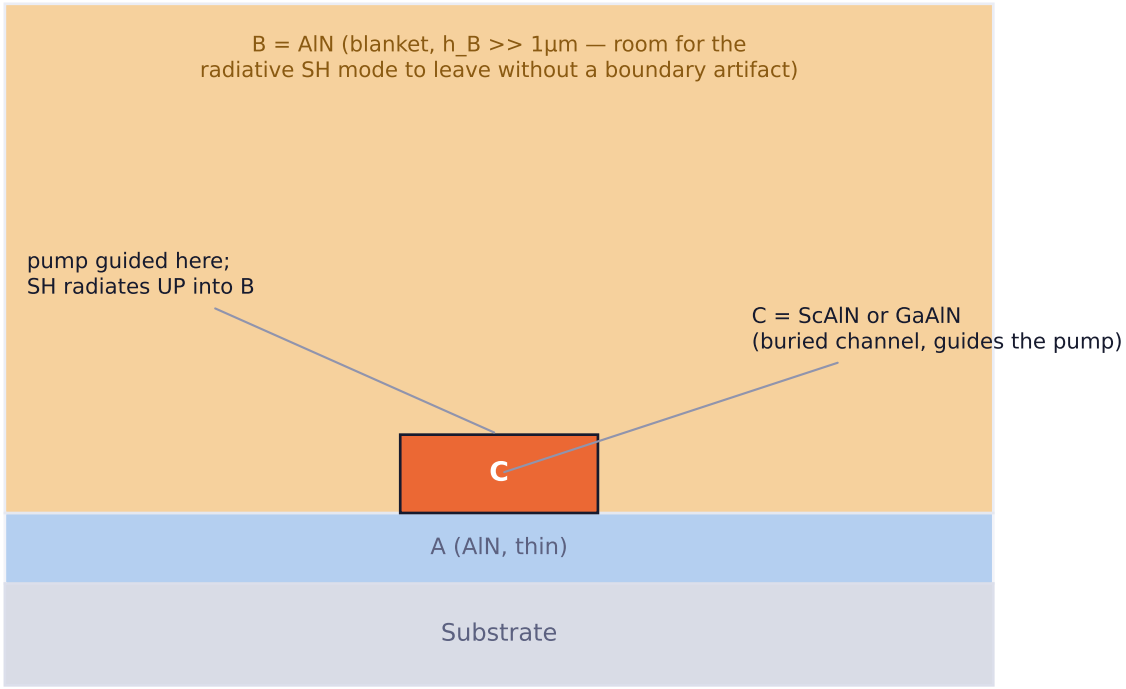
✓ **#7** AlN / ScAlN — both-active variant

Core C wrapped on the sides (and above) by shell B (w_C <= w_B) -- the corrected `hybrid` family, fixed 2026-10-10 after the earlier 'pedestal' geometry bug (core was stacked on top of the shell, not embedded in it -- see PLAN.md). Tests the Honda et al. 2025 TM00->TM20 mechanism; the re-run (row 6) is the next EMode campaign queued, launching with the standing sh_absorption_calibrated=True assumption from the start (already wired into both configs below). Row 7 (ScAlN shell) is ready too, using the 32%-Sc working d33 (section 6b of Input Assumptions, AT RISK).

SWEPT RANGE: lateral (`hybrid_sweep`) w_core 200-600 x sin_width 700-1500nm, fixed h_core=400/sin_height=700, 24 pts; vertical (`hybrid_vertical`) h_core 250-450 x sin_height 450-1100nm at the lateral sweep's best point, 23 pts (450 added to probe the pedestal data's "smaller wins" trend to the 200nm-margin boundary).

Topology 4: Embedded (Cerenkov)

SCHEMATIC — not yet built



MATRIX ROWS

- ○ **#8** ScAlN / AlN (blanket) — buried high-index channel radiating into thick AlN
- ○ **#9** GaAlN / AlN (blanket) — same, GaAlN channel

Same wrap-around embedding as Topology 3, but the material roles flip and B grows large: C (small, buried) carries the enhanced index/nonlinearity (ScAlN or GaAlN) and guides the pump; B becomes a blanket AlN cladding thick enough (>>1 um) to host a genuinely radiative SH continuum rather than a tightly confined guided mode. No new plumbing needed for the Cerenkov physics itself -- `cerenkov_fdfd.py`'s driven 2D solve already takes eps_fn/d_fn/bbox generically from any geometry class, the same interface every existing family implements. What's missing is the geometry class itself (not yet written) and, for row 9 specifically, a live transparency check -- GaAlN's favorable (higher-d33) direction is Ga-rich, but GaN's bandgap sits below the SH photon energy, so the useful window may be narrow or may not exist; not yet checked quantitatively (Input Assumptions report, section 5).

Sequencing

Proposed order, each step its own check-in (user direction, 2026-10-12: not a green light to run all ten rows unsupervised) -- every step launches with geom.params['sh_absorption_calibrated']=True from the start (page 1), not reprocessed later --

1. Row 6: re-run `hybrid1`/`hybrid_sweep1` (lateral, `survey_config_hybrid_sweep.py`) and `hybrid_vertical1` (vertical, `survey_config_hybrid_vertical.py`, now extended to sin_height=450) under the corrected, now-embedded geometry. Already queued, blocked only on this plan's review.
2. Row 4: SiN rib-loaded (`survey_config_loaded_sin.py`, 8 geometries, same grid as the TiO2 baseline), cheap, fills in topology 2 properly now that TiO2 (row 3) is known bad.
3. Rows 2/5/7: ScAlN d33 question resolved 2026-10-12 (Input Assumptions report section 6b) -- using the higher sputtered value (32% Sc, d33=54.6/d31=3.95 pm/V), explicitly flagged AT RISK pending the MBE disagreement. All three now ready to launch with no further blocker; configs not yet written (mirror row 1/4/6's grids with ScAlN swapped in).
4. Row 8: embedded-Cerenkov. d33 set (same AT RISK value), still needs the embedded-Cerenkov geometry class (topology 4) built -- doesn't exist yet, the remaining blocker.
5. Row 9: GaAlN. Needs n(lambda)/k(lambda) sourced first (Input Assumptions report, section 5) -- lowest priority given the bandgap/transparency concern, unless that check turns up favorable.
6. Row 10: substrate swap, cross-cutting -- can run alongside any of the above once the anisotropic-vs-isotropic AlN-substrate treatment is settled (user direction, 2026-10-12: anisotropic).